LESSON 9 • TERM 2 • LEVEL 1

# Picture Equations

A shape hides a number

**TODAY WE WILL**

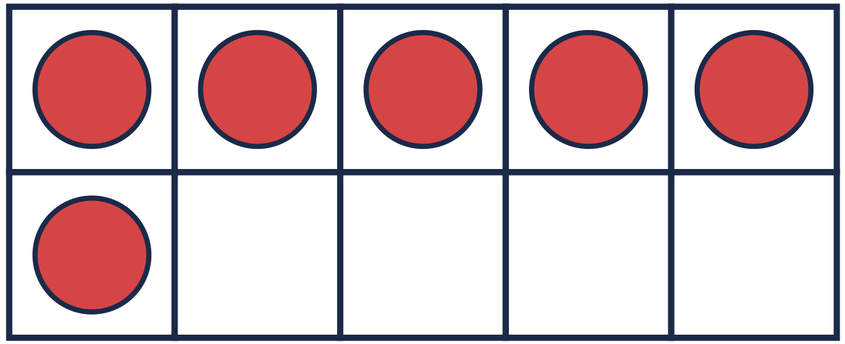
Find the number a shape is hiding.



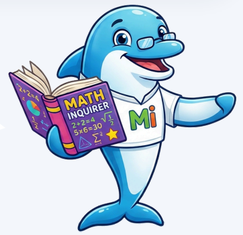
# Warm-up: a box hides a number

2–8 MIN

**Touch the counters. Say the number in the box.**



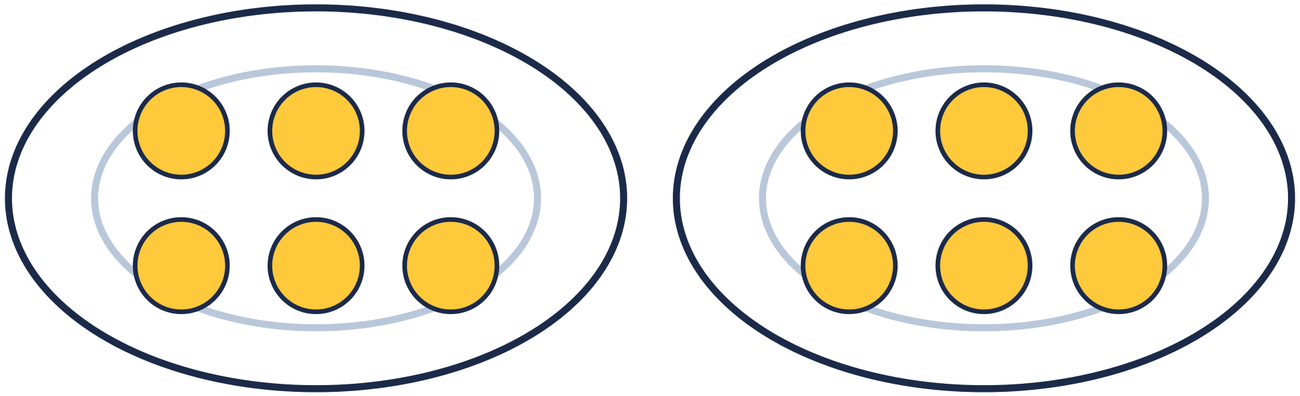
| | | | | |
|---|---|---|---|---|
| 6 | + | ☐ | = | 10 |
| ☐ | + | 3 | = | 8 |
| 9 | = | 5 | + | ☐ |




A box can hide a number!

# The same number twice

2–8 MIN

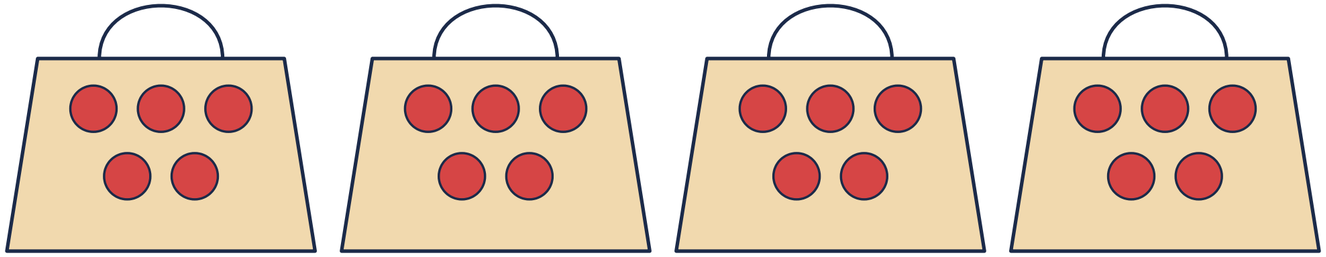





Two plates with the same number of counters.

# 4 groups of 5

2–8 MIN



| 4 | × | 5 | = | |
|---|---|---|---|---|

4 × 5 means 4 groups of 5.

# Sneak-peek puzzles

8–12 MIN

Ideas we have not learnt yet. A guess is a great start. Nobody is marked.

## PUZZLE 1: anyone can try

A cup hides some counters.

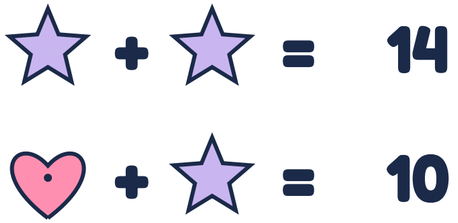
The cup and 4 loose counters make 9 in all.

How many counters are in the cup?

## PUZZLE 2: think

Each shape is a number.

How many is the heart?



# Explore: a shape and a pile

12–24 MIN



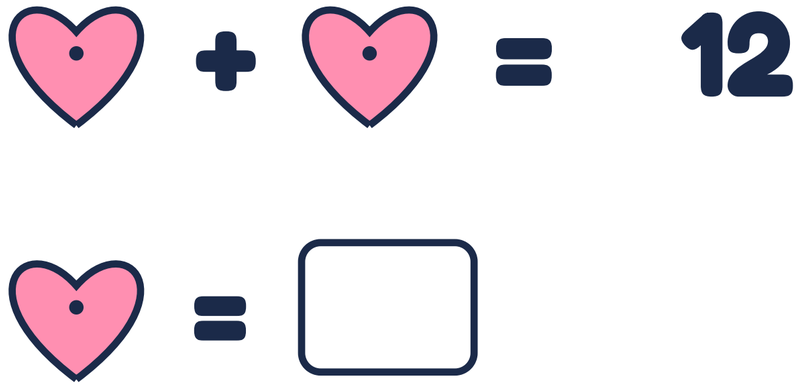




1. Count out 10 counters.
2. Put 4 on the square card.
3. The rest go on the circle card. How many?

# The same shape twice

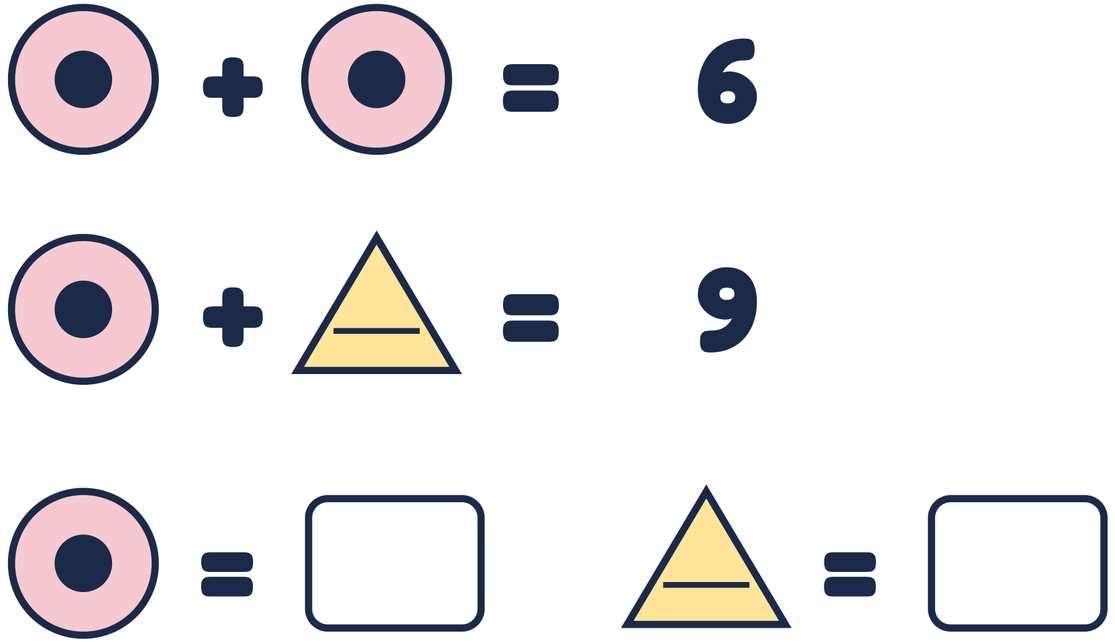
12–24 MIN




1. Count out 12 counters.
2. Share them onto 2 heart cards, one at a time.
3. The piles must match. How many on one heart?

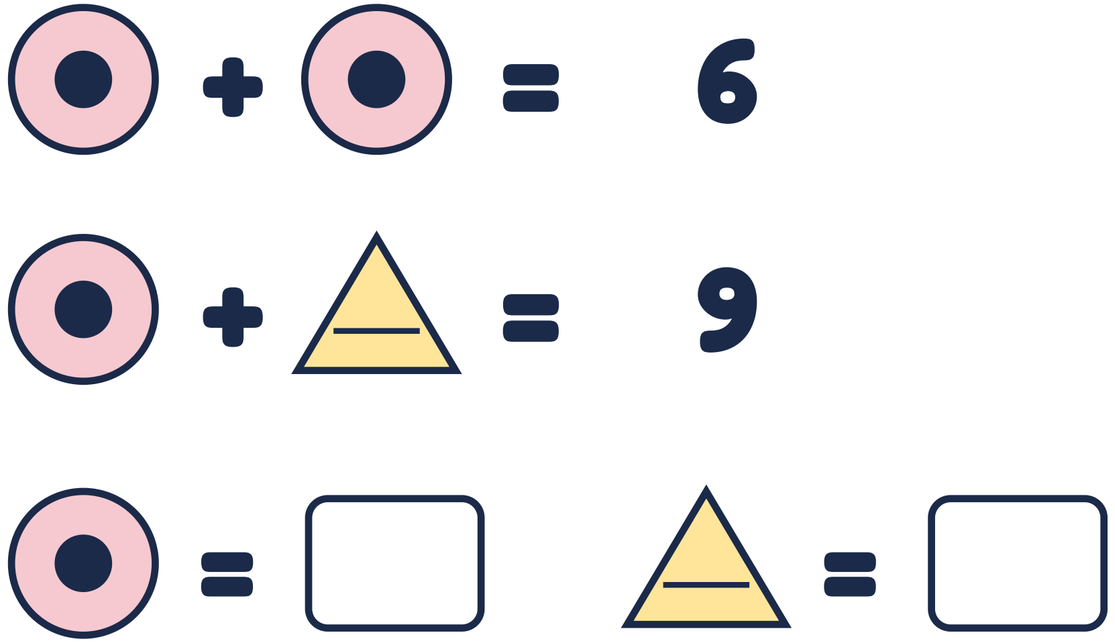
# Two lines, same shapes

12–24 MIN



# Two lines, same shapes

12–24 MIN




- Line 1 tells you the circle.
- The circle is the same number in line 2.

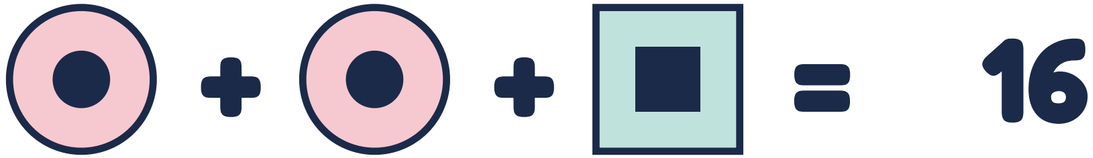
# Three rules

24–36 MIN

1. Each shape is a number.
2. The same shape is the same number every time.
3. In our puzzles, different shapes are different numbers.

# Find it, then check it

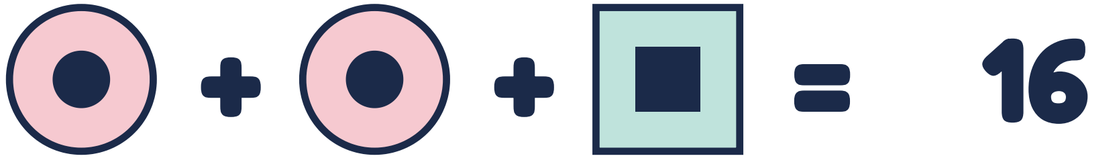
24–36 MIN





# Find it, then check it

24–36 MIN





**STEPS**

Take the square away from the total.

Share what is left between the 2 equal circles.

Check: put the number back into the line.

# Take one line from another

24–36 MIN





# Take one line from another

24–36 MIN







STEPS

Line 1 has one more star than line 2.

The totals show how much that extra star is.

Then put the star into line 2 to find the triangle.

24–36 MIN

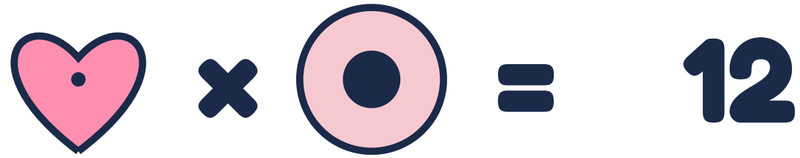
# Check in both lines

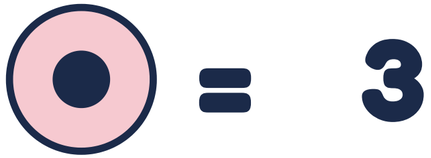
| Line | Check | Does it match? |
| --- | --- | --- |
| Line 1 | 5 + 5 + 7 = 17 | Yes |
| Line 2 | 5 + 7 = 12 | Yes |

Check in every line, not just one.

# A times line

24–36 MIN




STEPS

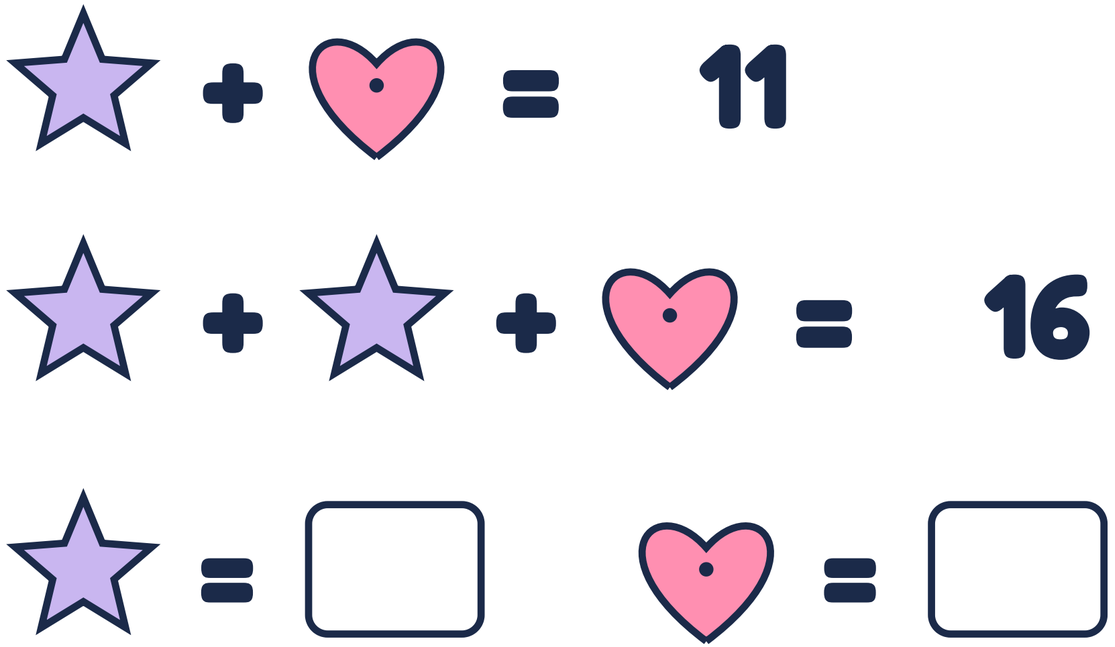
heart × circle means heart groups of the circle.

How many groups of the circle make the total?

Check: put the number back into the line.

# Your turn

24–36 MIN




**Find the star and the heart. Then check both lines.**

# Check: your turn

24–36 MIN

| Shape | Number | Check |
| --- | --- | --- |
| Star | 5 | Line 1: 5 + 6 = 11 |
| Heart | 6 | Line 2: 5 + 5 + 6 = 16 |

36–40 MIN

# Clap and whisper

Count to 20 with me. Clap on 5, 10, 15 and 20.

Whisper the numbers in between.

Stand up. Sip some water.

# Workbook page 1: Start here

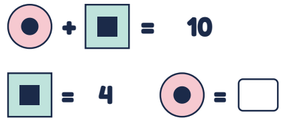
40–70 MIN

Questions 1 to 4. Take your time.

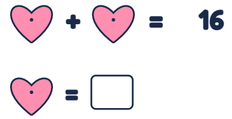
**S1** A cup hides some counters. 3 counters are outside the cup. There are 11 counters in all. Fill in the blank.



**S2** Each shape is a number. Find the number for the circle.




**S3** Each shape is a number. Find the number for the heart.




**S4** Each shape is a number. Find the number for the star.



# Check page 1

40–70 MIN

| | |
|---|---|
| S1 | 8 |
| S2 | 6 |
| S3 | 8 |
| S4 | 5 |

# Workbook page 2: Start here

40–70 MIN

Questions 5 to 7. Take your time.

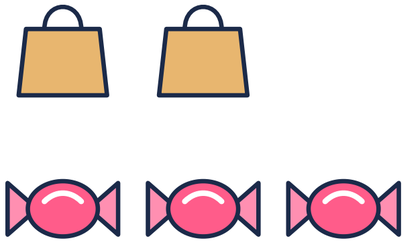
**S5** Each shape is a number. The square is 4. Fill in the blanks.




**S6** Ben says the circle is 7. Look at the picture.




**S7** Each bag has the same number of sweets. 3 sweets are outside the bags. There are 13 sweets in all. Fill in the blanks.



# Check page 2

40–70 MIN

| | |
|---|---|
| S5 | 10; 5 |
| S6 | No; 6 |
| S7 | 10; 5 |

40–70 MIN

# Choose your next page

| Page | What it is |
| --- | --- |
| **Think harder (page 3)** | Two lines and a times line |
| **Go further (page 4)** | Three shapes at once |
| **Investigation (page 5)** | One line or two? |
| **Check** | Finish and check your own work |

**Everyone may choose. Not finished? Pages 3 to 5 are for home too.**

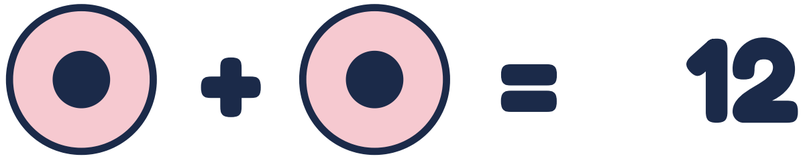
# Talk and explain

70–76 MIN

AISHA SAYS

Circle + circle = 12.

So circle + circle + circle = 18.




Is Aisha right? How could we check?

I think … because …

70–76 MIN

# Back to the sneak-peek



14



10



= 7



= 3

**The star is 7. The heart is 3. The cup holds 5.**

# Exit ticket

76–82 MIN

On your own. No looking back.

**E1** **Each shape is a number. The triangle + the triangle = 18. Find the number for the triangle.**

The triangle is ▢.

**E2** **The heart + the square = 13. The square is 5. Find the number for the heart. Choose the correct answer. Write its number in the bracket.**

**E3** **Each shape is a number. The star is 5. Find the number for each circle.**




Each circle is ▢.

# Check the exit ticket

76–82 MIN

| | |
|---|---|
| E1 | 9 |
| E2 | (4) 8 |
| E3 | 7 |

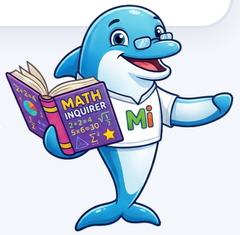
# Look-back

82–85 MIN

**Three minutes. There are no right answers. You can pass. Nobody reads it out.**

| | |
|---|---|
| **Something I did well** | write or draw one thing |
| **The hardest part was …** | ☐ finding the number for one shape ☐ using two lines<br>☐ checking in every line ☐ something else |
| **What helped me** | ☐ counters and shape cards ☐ drawing it ☐ checking my answer<br>☐ I was not stuck |
| **Next time I will try** | ☐ using counters ☐ checking my answer ☐ drawing it<br>☐ asking a question |
| | ☐ I am still stuck (my grown-up will photograph this page for my tutor)<br>☐ I would like to pass |

**Take your time. You can pass.**



85–90 MIN

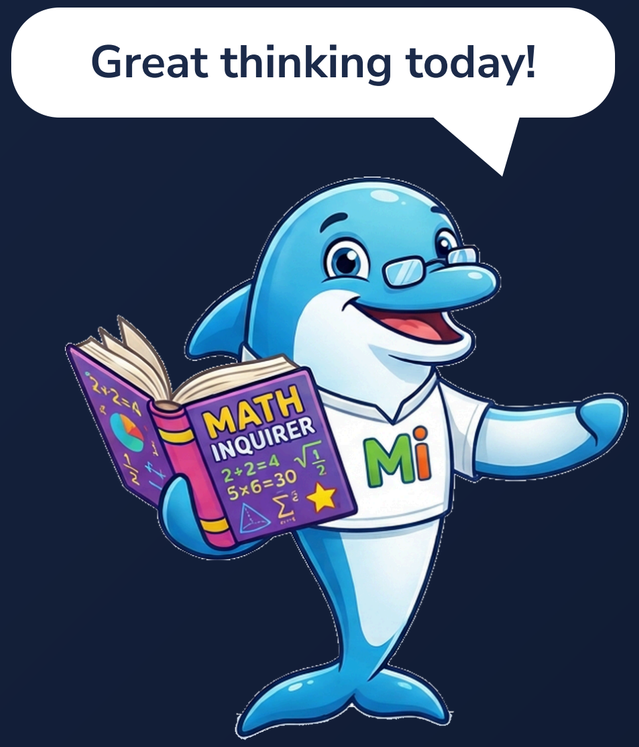
# Well done, inquirers!

Show a grown-up one thing you found out today.

NEXT LESSON

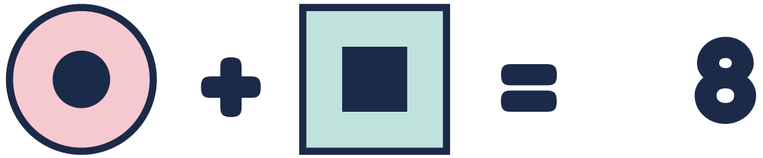
Find One First, and Mi's Challenge Quest

Have ready: the Term 2 Lesson 10 workbook with its tokens page cut out (30 tokens), 3 to 6 small things, the Quest Log in its envelope (keep it closed), a quest partner, a pencil and paper.



# Go further: one line or two?

OPTIONAL




Could the circle be 2? Could it be 3?




What does the second line tell us?

OPTIONAL

# If we have time: pick one

| Activity | What we do |
| --- | --- |
| **Can a shape change?** | Ali says the heart is 5 in one line and 6 in another. |
| **Make a puzzle** | Pick two numbers. Write two lines for your grown-up. |
| **Teach it back** | Show a toy how a cup hides a number |

**Optional. Everyone may join.**